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A new metric for determining the importance of transient storage

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Abstract. A review of various metrics used to characterize transient storage indicates that none of the existing measures successfully integrate the interaction between advective velocity and the transient storage parameters (storage zone area, storage zone exchange coefficient). Further, 2 existing metrics are related to mean travel time, a quantity that is independent of the storage zone exchange coefficient, α. This interaction and the effect of α on travel time are important considerations when determining the mass of solute entering the storage zone within a given reach. A new metric based on median reach travel time is therefore proposed. Median reach travel time due to advection–dispersion and transient storage, and median reach travel time due solely to advection–dispersion are computed based on numerical simulations. These 2 travel times are used to determine F_{med}, the fraction of median travel time due to transient storage. Application of the new metric to 53 existing parameter sets indicates that transient storage accounts for 0.12% to 68.0% of total reach travel time. Rankings of storage zone importance based on the new metric are substantially different from rankings based on storage zone residence time, storage exchange flux, and the hydrological retention factor. These differences result from the ability of the new metric to characterize the interaction between advective velocity and transient storage, and the resultant effects on reach travel time and mass transport.

Key words: transient storage, hyporheic zone, solute transport, travel time, tracer, OTIS.

Transient storage has been observed in many streams and small rivers where solutes are temporarily detained in the hyporheic zone and surface features (small eddies and pools) with low longitudinal velocities. Because transient storage acts to delay the downstream transport of solute mass, it has important implications for nutrient cycling and contaminant transport in stream ecosystems. A mathematical model that considers the effects of transient storage on mass transport has been developed (Bencala and Walters 1983, Hart 1995, Runkel 1998). This modeling approach has been used extensively in recent years to quantify hydrodynamic and biogeochemical processes (Bencala 1984, Bencala et al. 1990, Stream Solute Workshop 1990, Broshears et al. 1993, D'Angelo et al. 1993, Harvey and Bencala 1993, Valett et al. 1996, Morrice et al. 1997, Mulholland et al. 1997, Runkel et al. 1998, Hart et al. 1999). Additional research efforts have focused on analysis of the transport equations (Runkel and Chapra 1993, Schmid 1995, Lees et al. 2000) as well as interpretation of model parameters (Harvey et al. 1996, Wagner and Harvey 1997).

Interpretation of model parameters is an important task as investigators attempt to relate the relevant parameters to physical stream characteristics. Stream characteristics of importance include advective velocity, storage zone size, and storage zone exchange. These stream characteristics are in turn used to infer information related to the location, timing, and magnitude of biogeochemical processes. Because of the growing interest in the effects of transient storage, several investigators have proposed metrics that may be used for intra- and interstream comparisons. A review of various metrics used to characterize transient storage indicates that none of the existing measures successfully integrates the interaction between advective velocity and the transient storage parameters (storage zone area; storage zone exchange coefficient, α). Further, two existing metrics are related to mean travel time, a quantity that is independent of α. The interaction between advective velocity and the transient storage parameters and the effect of α on median travel time are important considerations when determining the mass of solute entering the storage zone within a given reach. In light of this finding, a new metric for transient storage is developed.

Model Equations and Existing Metrics

The analyses presented are based on the transient storage equations as implemented within the OTIS solute transport model (Runkel 1998).

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These equations are functionally equivalent to alternate formulations presented in the literature (e.g., Thackston and Schnelle 1970, Nordin and Troutman 1980, Hart 1995). The equations describe the physical processes of advection, dispersion, and transient storage. Two conceptual areas are defined within the model: the main channel and the storage zone. The main channel is defined as the portion of the stream in which advection and dispersion are the dominant transport mechanisms. The storage zone is defined as the portion of the stream that contributes to transient storage, i.e., the hyporheic zone, pools, and eddies. The exchange of solute mass between the main channel and the storage zone is modeled as a 1st-order mass transfer process. Given this conceptual framework, equations describing the spatial and temporal variation in solute concentrations are given by:

$$\frac{\partial C}{\partial t} = -\frac{Q}{A}\frac{\partial C}{\partial x} + \frac{1}{A}\frac{\partial}{\partial x}\left(AD\frac{\partial C}{\partial x}\right) + \frac{q_L}{A}(C_L - C) + \alpha(C_S - C) \quad (1)$$

$$\frac{dC_S}{dt} = \alpha\frac{A}{A_S}(C - C_S) \quad (2)$$

where A is the main channel cross-sectional area (m^2), A_S is the cross-sectional area of the storage zone (m^2), C is the main channel solute concentration (mg/L), C_S is the storage zone solute concentration (mg/L), C_L is the lateral inflow solute concentration (mg/L), D is the dispersion coefficient (m^2/s), Q is the volumetric flow rate (m^3/s), q_L is the lateral inflow rate on a per length basis (m^3/s-m), t is time (s), x is distance (m), and α is the storage zone exchange coefficient (/s).

The model parameters presented above have been used to develop various metrics for intra- and interstream comparisons. A commonly used metric is the simple ratio of storage zone cross-sectional area and main channel cross-sectional area, A_S/A. Another common metric is L_s, the average distance a molecule travels downstream within the main channel prior to entering the storage zone (Mulholland et al. 1994):

$$L_s = \frac{u}{\alpha} \quad (3)$$

where u is advective velocity (Q/A) and Q is average flow in the reach. Division of L_s by u yields the average time a molecule remains in the main channel before passing into the storage zone:

$$T_{str} = \frac{1}{\alpha} \quad (4)$$

where T_{str} is the main channel residence time. After travelling L_s meters, the average molecule will remain in storage for a time given by:

$$T_{sto} = \frac{A_S}{\alpha A} \quad (5)$$

where T_{sto} is the storage zone residence time (Thackston and Schnelle 1970).

Two additional metrics include the storage exchange flux (Harvey et al. 1996) and the hydrological retention factor (Morrice et al. 1997). The storage exchange flux is equal to:

$$q_s = \alpha A \quad (6)$$

where q_s is the average water flux through the storage zone per unit length (Harvey et al. 1996). The hydrological retention factor is given by:

$$R_h = \frac{T_{sto}}{L_s} = \frac{A_S}{Q} \quad (7)$$

As shown in this paper, the effect of transient storage on solute mass is influenced by advective velocity (u) and the transient storage parameters A_S and α. None of the metrics presented above describe the overall effect of these 3 parameters on downstream transport. To complicate matters further, changes in parameter values may suggest increased or decreased importance of transient storage, depending on the metric considered. Decreased values of α, for example, lead to an increase in T_{sto}. This increase in T_{sto} may be interpreted to indicate that transient storage is more important in a given reach. In contrast, this same decrease in α will act to increase T_{str} and decrease q_s. These changes in T_{str} and q_s suggest a decreased importance of transient storage. Finally, R_h will be unaffected by the change in α. These conflicting interpretations of transient storage underscore the need for a unified metric. To this end, a new metric that considers the interaction between u, A_S, and α is presented.

Development of a New Metric

Mean travel time

A potential basis for a new metric is mean travel time. Mean travel time is given by (Nordin and Troutman 1980, Schmid 1995):

$$t_{mean} = \frac{Lu + 2D}{u^2} + \frac{A_S}{A}\left(\frac{Lu + 2D}{u^2}\right) \quad (8)$$

where L is reach length (m). The terms on the right-hand side of equation 8 are the portions of mean travel time due to the main channel ($t_{mean}{}^m$) and the storage zone ($t_{mean}{}^s$), respectively. A simpler equation for mean travel time may be developed by neglecting dispersion. In this case, mean travel time is equal to volumetric residence time (Thackston and Schnelle 1970):

$$t_{vol} = \frac{L}{u} + \frac{LA_S}{Q} \quad (9)$$

where the terms on the right-hand side are equal to volume/flow for the main channel ($t_{vol}{}^m$) and the storage zone ($t_{vol}{}^s$), respectively. For the special case of $L = L_s$, volumetric residence time is equal to the sum of the main channel and storage zone residence times (combining equations 3 and 9 yields $t_{vol} = T_{str} + T_{sto}$).

One approach to determining the overall effect of transient storage on the downstream transport of solutes is to consider the fraction of mean travel time that is due to transient storage. This fraction is equal to the 2nd term on the right-hand side of equation 8 divided by the entire right-hand side ($t_{mean}{}^s/t_{mean}$). After algebraic manipulation, this quantity is simply equal to the fraction of total reach volume occupied by the storage zone:

$$F_{mean} = \frac{A_S}{A + A_S} \quad (10)$$

An identical relationship exists for the case of t_{vol} (equation 9) and the residence time metrics defined previously:

$$\frac{T_{sto}}{T_{str} + T_{sto}} = \frac{A_S}{A + A_S} \quad (11)$$

Note that this fraction is similar in form to the commonly used metric, A_S/A. Use of F_{mean} may be preferable to A_S/A because of its basis in theory (equations 8 and 9) and the fact that it is bounded between 0 and 1.

Although the results presented as equations 10 and 11 are of interest, the primary goal of developing a new metric has not been realized because F_{mean} contains only 1 of the 3 important parameters, A_S. The failure of F_{mean} to include the effects of α and u arises from the fact that velocity contributes to both the main channel and storage zone portions of t_{mean} (equation 8). Further, t_{mean} is independent of α.

The independence of t_{mean} from α is illustrated in Fig. 1 where an instantaneous slug injection is considered. The solid line depicts a simulation based on reach 3 of Little Lost Man Creek (Bencala 1984). Application of the transient storage model yields a skewed concentration versus time profile, such that t_{mean} occurs long after the time of peak concentration. A 2nd simulation in which α is decreased by an order of magnitude is shown as a dotted line. This decrease in α causes an increase in L_s, such that fewer molecules enter the storage zone over the experimental reach. Because the molecules that remain in the main channel are subject to advection, the decrease in α (increase in L_s) acts to shift more solute mass to the left; i.e., most tracer molecules have shorter travel times. This shift in mass is counterbalanced by the longer travel times associated with the molecules that make up the tail of the tracer profile; i.e., although fewer molecules enter the storage zone, they remain within the storage zone for a longer period of time (T_{sto} increases). (Note that the tail of the 2nd simulation exceeds that of the 1st for all times $\geq$27 h (not shown in Fig. 1)). The net effect of the change in α on t_{mean} and t_{vol} is therefore nil (Fig. 1).

Median travel time

In the example provided above, a relatively small number of tracer molecules with extremely long travel times act to skew the travel time distribution to the right, such that t_{mean} is unaffected by a decrease in α. This independence of travel time on α can be eliminated by considering median travel time, as the median is unaffected by extreme values. Unlike t_{mean}, an analytic expression such as that provided by equation 8 is not available for the median (B. H. Schmid, Technische Universität Wien, Vienna, Austria, personal communication). The fraction of median travel time due to transient storage is therefore determined based on the numerical

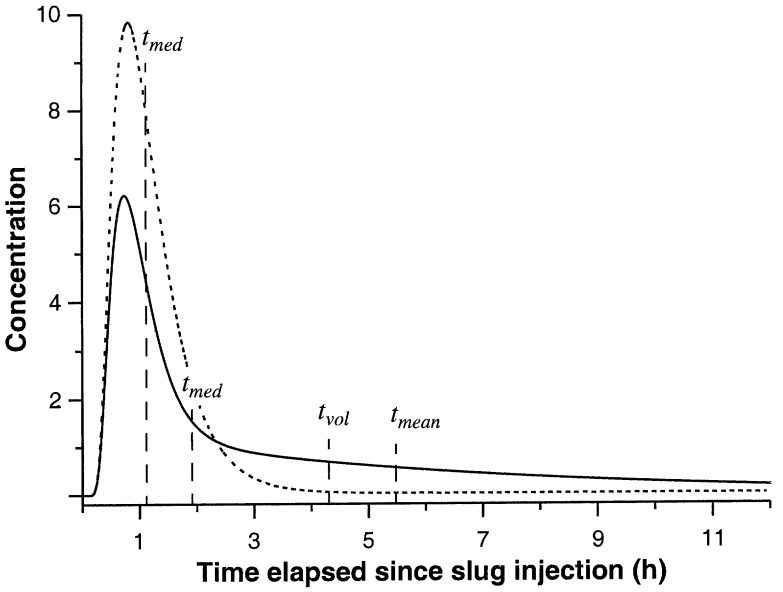


FIG. 1. Time versus concentration profiles resulting from a slug injection, showing mean travel time (t_{mean}), volumetric residence time (t_{vol}), and median travel time (t_{med}). Solid line is the simulated concentration based on parameters from the 3rd reach of Little Lost Man Creek, California (Bencala 1984); dotted line is the simulated concentration based on the same parameters, but with the storage zone exchange coefficient, α, reduced by an order of magnitude.

solution of equations 1 and 2 as follows. First, median travel time is the time at which:

$$\frac{\int_0^t QC(x = L, t)\, dt}{\int_0^\infty QC(x = L, t)\, dt} = 0.5 \qquad (12)$$

where C is the concentration that results from an instantaneous slug injection. The fraction of median travel time due to transient storage is then given by:

$$F_{med} = \frac{t_{med} - t_{med}{}^m}{t_{med}} = \frac{t_{med}{}^s}{t_{med}} \qquad (13)$$

where t_{med} is the total median travel time, $t_{med}{}^m$ is the median travel time due to the main channel, and $t_{med}{}^s$ is the median travel time due to the storage zone. As given by equation 12, median travel time corresponds to the center of mass, a characteristic of the tracer profiles that is affected by α (Fig. 1).

Calculation of t_{med} and $t_{med}{}^m$ for use in equation 13 is illustrated in Fig. 2. For the case of steady flow, Q drops out of equation 12 and the area under the time versus concentration plot is proportional to solute mass. The relative amount of solute mass may therefore be determined by numerical integration of the concentration profile (Fig. 2A). The time at which ½ the mass passes reach length L is equal to the median travel time. Calculation of the total median travel time (t_{med}) is based on the concentration profile resulting from the solution of equations 1 and 2 (Fig. 2B); calculation of the median travel time for the main channel ($t_{med}{}^m$) is based on the concentration profile that occurs in the absence of transient storage (advection and dispersion only, equation 1 with $\alpha = 0$) (Fig. 2B). The fraction of median travel time due to transient storage (F_{med}) is then determined using equation 13.

Although it is possible to determine F_{med} using the procedure outlined above (equation 12, Fig. 2), it is somewhat problematic because of the lack of a slug boundary condition within the OTIS solute transport model. An alternate procedure based on the time to plateau for a continuous injection is therefore described below. The equivalence of the 2 procedures may be seen by considering the relationship between median travel time and the time required to reach ½ of the plateau concentration; because

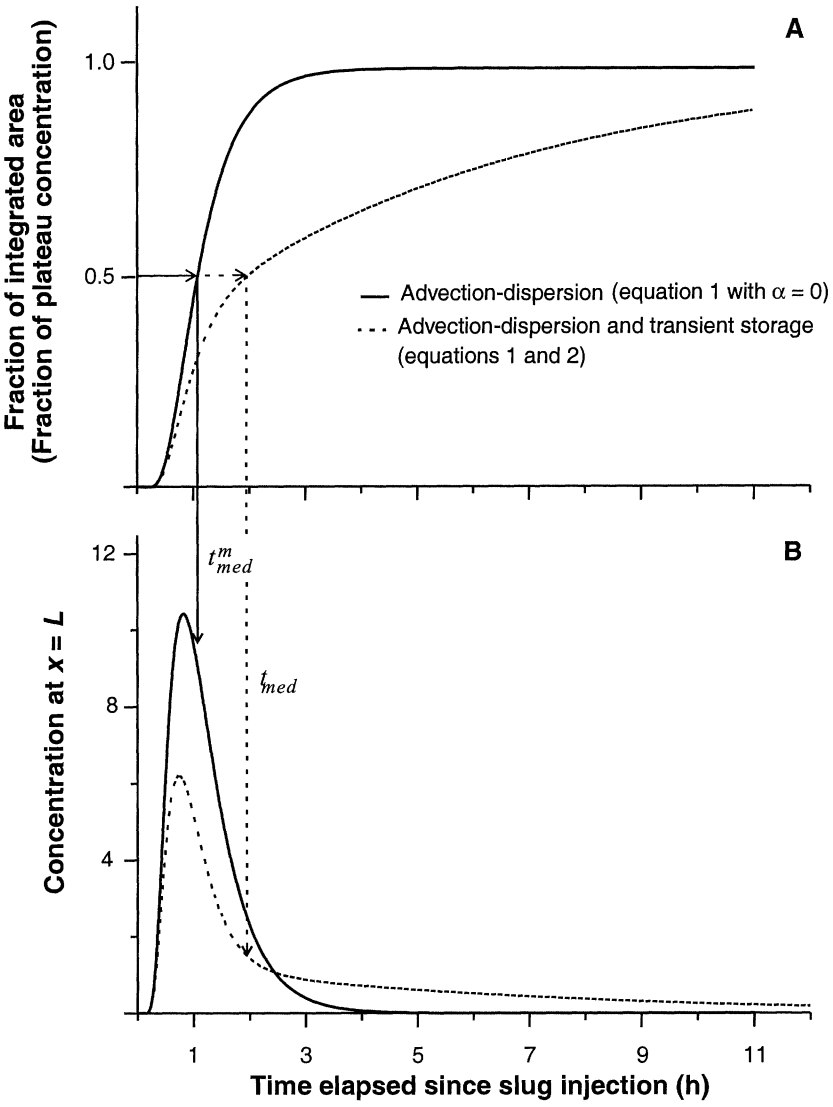


FIG. 2. Calculation of median travel times based on integration of the concentration profile that results from a slug injection. A.—Median travel time is equal to the time at which ½ of the integrated area is realized. B.—Total median travel time (t_{med}) is calculated using the concentration profile that results from advection–dispersion and transient storage; median travel time due to the main channel ($t_{med}{}^m$) is calculated using the concentration profile that results from advection–dispersion without transient storage. x = distance, L = reach length.

median travel time corresponds to the time when ½ the mass has passed the observation point, median travel time is equivalent to the time at which ½ of the plateau concentration is realized. This equivalence can be seen by noting that the integrated area for a slug injection is identical to the concentration profile for a continuous injection, i.e., a continuous injection results in concentration profiles that are identical to the curves shown in Fig. 2A, where the ordinates correspond to the fraction of plateau concentration. F_{med} may therefore be determined as follows:

1) t_{med}.—A transport model is used to simulate the effects of a continuous injection in the presence of transient storage (equations 1 and 2), with concentrations at the reach end-

TABLE 1. Streams included in analysis of median travel time due to storage.

Stream	Reference	Number of parameter sets	Abbreviation	Notes
Coweeta, North Carolina	D'Angelo et al. 1993	4	Cow-#	# in abbreviation refers to the order of the gradient sites (see table 2 of reference)
Gallina Creek, New Mexico	Morrice et al. 1997	4	Gall-xxx	xxx in abbreviation refers to season (see table IV of reference)
Hugh White Creek, North Carolina	Mulholland et al. 1997	1	HWC	
Little Lost Man Creek, California	Bencala 1984	4	LLM-#	# in abbreviation refers to reach number (see table 3 of reference)
Snake River, Colorado	Bencala et al. 1990	8	Snake-#	# in abbreviation refers to reach number (see table 2b of reference)
St. Kevin Gulch, Colorado	Broshears et al. 1993	4	SK86-#	# in abbreviation refers to reach number (see table 3 of reference)
St. Kevin Gulch, Colorado	Broshears et al. 1996	4	SK88-#	# in abbreviation refers to reach number (see table 5 of reference)
Uvas Creek, California	Bencala and Walters 1983	3	Uvas-#	# in abbreviation refers to reach number (see table 1 of reference)
West Fork Walker Branch, Tennessee	Mulholland et al. 1997	1	WFWB	
West Fork Walker Branch, Tennessee	Hart et al. 1999	20	WFWB-#	# in abbreviation refers to the study listed in table I of reference

point output at 0.001-h intervals. The time required to reach ½ of the plateau concentration is determined by interpolation between the time–concentration points. Total median travel time (t_{med}) is set equal to the interpolated value, minus the time at which the injection is initiated.

2) t_{med}^{m}.—A transport model is used to simulate the effects of a continuous injection in the absence of transient storage (equation 1 with $\alpha = 0$), with concentrations at the reach endpoint output at 0.001-h intervals. The time required to reach ½ of the plateau concentration is determined by interpolation between the time–concentration points. Median travel time due to the main channel (t_{med}^{m}) is set equal to the interpolated value, minus the time at which the injection is initiated.

3) F_{med}.—Values of t_{med} and t_{med}^{m} determined in steps 1 and 2 are used in equation 13 to determine F_{med}.

Application

Use of median travel time to quantify the effects of transient storage is demonstrated by considering 53 parameter sets obtained from the published literature (Table 1, Appendix). As with other measures of solute transport and uptake (e.g., Essington and Carpenter 2000), median travel time and F_{med} are scale-dependent quantities. Values of F_{med} were therefore computed for various multiples of the average distance travelled, L_s, to evaluate length dependence (Fig. 3). Parameter sets with spatially varying flow ($q_L > 0$) were excluded from this analysis because of the effect of distance on velocity (u increases with x, such that L_s becomes

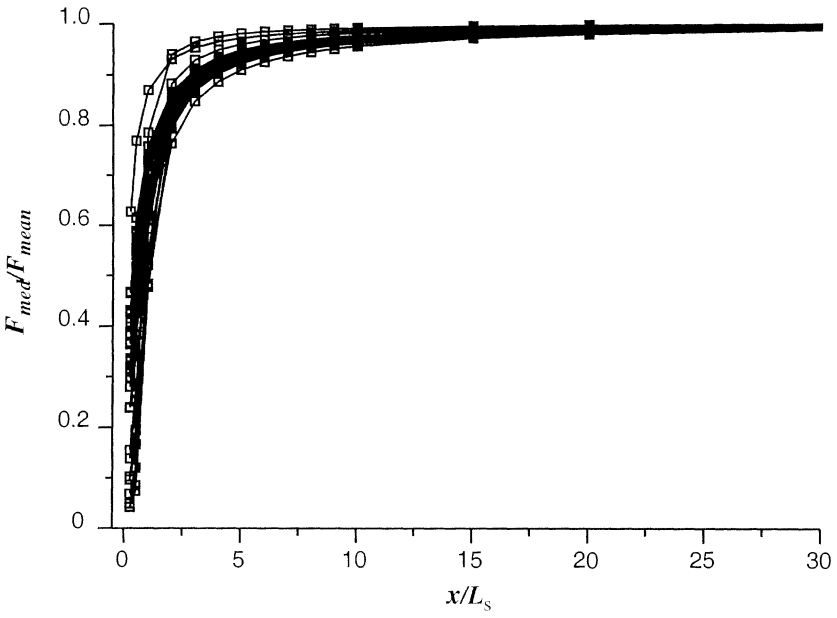


FIG. 3. Relationship between median travel time due to storage and length, for the 32 parameter sets with $q_L = 0$. The fraction of median travel time due to storage (F_{med}) approaches the fraction of mean travel time due to storage (F_{mean}) as x/L_s increases. x = distance, L_s = average distance travelled prior to entering storage zone, q_L = lateral inflow rate.

a moving target). As shown in Fig. 3, F_{med} approaches F_{mean} as x/L_s increases (i.e., the mean and median converge as the number of times the storage zone is sampled increases).

The strong length dependence of F_{med} presents a challenge for the development of a new metric based on median travel time, in that comparison of parameter sets from studies with various reach lengths must be evaluated at some standard distance. Determination of the standard distance was made by keeping in mind the original goal, i.e., to develop a new metric that considers the interaction of u, A_s, and α. Inspection of Fig. 3 indicates that long distances tend to weight F_{med} in favor of A_s (for $x/L_s > 5$, $F_{med} \simeq F_{mean}$) , whereas short distances tend to favor u and α (for $x/L_s < 2$, F_{med} is sensitive to changes in L_s). Appropriate weighting between the effects of F_{mean} and L_s was therefore determined by evaluating F_{med} for the 53 parameter sets at various values of x. For each value of x, the parameter sets were ranked according to F_{med}, F_{mean}, and L_s. A value of x = 200 m provides approximately equal weighting of F_{mean} and L_s (Table 2) and is proposed as the standard distance.

TABLE 2. Criteria used to establish a standard distance for the fraction of median travel time due to transient storage (F_{med}). F_{mean} = fraction of mean travel time due to storage, L_S = average distance travelled prior to entering storage zone.

Distance for evaluation of F_{med} (m)	Average rank of parameter sets appearing in the top 10 parameter sets according to F_{med}, if parameter sets were ranked according to:	
	L_S	F_{mean}
100	7.5	14.9
200	11	10.4
300	13.3	9.2

Evaluation of F_{med} at the standard distance (F_{med}^{200}) indicates that transient storage accounts for 0.12 to 68% of total reach travel time for the 53 parameter sets considered (Table 3). The relationship between F_{med} and the model parameters may be seen by noting the asymptotic behavior of F_{med}/F_{mean} with respect to x/L_s (Fig. 3). This asymptotic behavior suggests a functional relationship of the form:

$$F_{med} \cong (1 - e^{-L(\alpha/u)}) \frac{A_S}{A + A_S} \qquad (14)$$

This relationship for the case of L = 200 m

TABLE 3. The 53 parameter sets ranked according to the fraction of median travel time due to storage, evaluation at 200 m (F_{med}^{200}). F_{mean} = fraction of mean travel time due to storage, L_S = average distance travelled prior to entering storage zone, R_h = hydrological retention factor, T_{sto} = storage zone residence time, q_s = storage exchange flux. Abbreviations of reaches as in Table 1.

	F_{med}^{200} (%)	F_{mean} (%)		L_S (m)		R_h (s/m)		q_s (m²/s)		T_{sto} (min)	
Reach	Value	Value	Rank	Value	Rank	Value	Rank	Value	Rank	Value	Rank
LLM-3	68.01	71.43	4	64.2	1	194.68	3	1.20×10^{-4}	11	208.33	14
LLM-2	62.40	80.00	2	188.0	12	265.96	2	4.00×10^{-5}	20	833.33	5
HWC	49.33	59.72	8	108.5	2	23.30	10	5.10×10^{-5}	17	42.12	21
Gall-Aut	47.49	82.14	1	245.9	17	306.67	1	3.05×10^{-6}	50	1256.83	3
Gall-Sum	37.83	61.90	7	208.3	13	65.00	6	9.60×10^{-6}	41	225.69	12
Cow-2b	23.01	29.47	18	119.7	3	2.91	27	4.60×10^{-4}	4	5.80	46
Gall-Win	19.69	41.67	14	214.3	14	6.67	17	7.00×10^{-5}	14	23.81	24
Cow-2	16.28	23.86	22	174.1	10	2.00	30	3.45×10^{-4}	5	5.80	45
Cow-4	14.97	19.49	25	127.2	5	1.27	40	2.17×10^{-3}	1	2.69	50
Uvas-5	12.91	75.00	3	589.7	32	113.04	4	2.34×10^{-5}	26	1111.11	4
Snake-3	12.70	44.12	13	349.6	22	4.52	21	9.50×10^{-4}	3	26.32	23
WFWB-6	10.38	13.33	29	145.1	8	1.82	32	7.58×10^{-5}	12	4.40	49
WFWB-14	9.79	14.73	26	174.5	11	5.94	19	1.83×10^{-5}	29	17.27	30
WFWB-13	9.23	11.29	33	120.8	4	4.52	22	2.57×10^{-5}	24	9.09	40
LLM-4	9.09	50.00	9	649.2	34	77.02	5	1.20×10^{-5}	37	833.33	6
WFWB-4	8.80	11.21	34	161.9	9	0.60	45	2.47×10^{-4}	7	1.62	51
LLM-1	8.36	25.00	21	509.4	28	16.36	12	1.44×10^{-5}	36	138.89	15
Cow-3	8.15	12.27	30	136.4	7	0.51	47	1.14×10^{-3}	2	1.17	52
WFWB-8	7.42	12.00	32	218.2	16	4.69	20	1.47×10^{-5}	35	17.05	31
WFWB-18	6.82	9.91	39	217.9	15	1.59	36	3.17×10^{-5}	21	5.79	47
WFWB-7	6.61	12.09	31	353.0	23	1.38	39	4.53×10^{-5}	18	8.09	44
WFWB-2	6.30	9.72	40	300.0	20	1.08	42	4.33×10^{-5}	19	5.39	48
WFWB-3	6.25	6.86	50	133.0	6	0.35	49	3.01×10^{-4}	6	0.78	53
WFWB-15	6.24	10.57	36	290.9	19	4.06	23	1.10×10^{-5}	38	19.70	28
WFWB-20	5.63	7.89	47	250.9	18	2.93	26	1.63×10^{-5}	31	12.24	35
WFWB-1	5.35	8.40	42	361.1	25	1.69	34	1.80×10^{-5}	30	10.19	37
WFWB-5	5.32	11.11	35	400.0	26	1.25	41	3.00×10^{-5}	22	8.33	43
SK88-2	5.03	38.17	15	511.0	29	12.08	14	2.43×10^{-5}	25	102.88	18
WFWB-19	4.89	8.16	43	359.9	24	1.48	37	1.50×10^{-5}	33	8.89	42
WFWB-9	4.59	10.26	37	445.7	27	3.08	25	5.83×10^{-6}	45	22.86	25
Uvas-3	4.44	50.00	10	1194.4	40	27.91	8	1.08×10^{-5}	39	555.56	8
WFWB-16	4.31	6.98	49	318.8	21	1.76	33	1.07×10^{-5}	40	9.38	38
WFWB-12	3.78	7.41	48	533.3	30	1.00	43	2.25×10^{-5}	27	8.89	41
WFWB-10	3.53	8.16	44	560.0	31	1.90	31	7.50×10^{-6}	42	17.78	29
SK88-1	3.50	44.78	12	664.9	35	12.20	13	1.85×10^{-5}	28	135.14	16
WFWB	3.07	8.57	41	640.6	33	2.09	28	6.72×10^{-6}	43	22.32	26
WFWB-11	2.93	8.16	45	833.3	36	1.60	35	6.00×10^{-6}	44	22.22	27
Gall-Spr	2.69	10.00	38	1041.7	39	0.53	46	7.20×10^{-5}	13	9.26	39
WFWB-17	2.58	8.05	46	850.0	37	2.06	29	4.00×10^{-6}	47	29.17	22
SK88-3	2.33	70.14	5	862.4	38	27.24	9	1.49×10^{-5}	34	391.50	10
Uvas-4	1.78	50.00	11	3280.5	44	30.48	7	4.10×10^{-6}	46	1666.67	1
Snake-4	1.19	21.74	24	1942.2	42	1.43	38	1.80×10^{-4}	8	46.30	20
SK86-1	0.90	29.41	19	1719.4	41	8.08	15	3.60×10^{-6}	49	231.48	11
Snake-5	0.58	14.63	27	2689.0	43	0.32	50	1.40×10^{-4}	10	14.29	33
SK88-4	0.50	27.78	20	3376.9	45	3.80	24	3.90×10^{-6}	48	213.68	13
SK86-2	0.40	34.01	16	3636.9	46	7.09	16	1.94×10^{-6}	53	429.55	9
Snake-9	0.37	23.81	23	4658.1	48	0.67	44	1.60×10^{-4}	9	52.08	19
Snake-2	0.24	3.51	53	10,672.2	50	0.07	53	2.75×10^{-5}	23	12.12	36
SK86-4	0.24	33.44	17	5343.5	49	6.27	18	2.99×10^{-6}	51	558.35	7
Snake-1	0.21	14.08	28	15,965.1	53	0.41	48	1.53×10^{-5}	32	109.29	17
SK86-3	0.17	64.43	6	4170.7	47	21.72	11	2.76×10^{-6}	52	1509.66	2
Snake-8	0.13	3.97	52	11,072.5	51	0.07	52	6.05×10^{-5}	15	13.77	34
Snake-7	0.12	4.42	51	11,673.2	52	0.08	51	5.40×10^{-5}	16	15.43	32

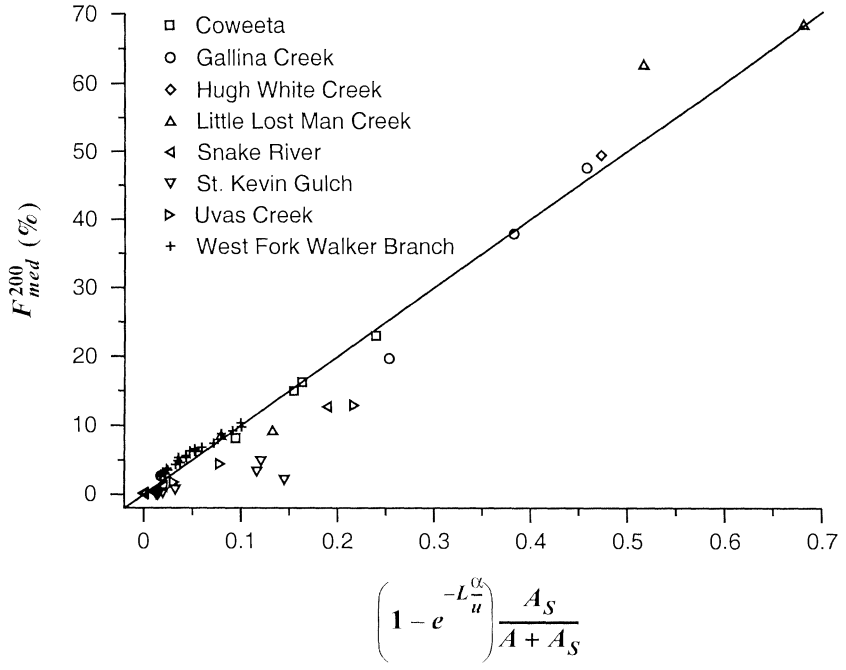


FIG. 4. Relationship between the fraction of median travel time due to storage, evaluated at 200 m (F_{med}^{200}) and the transient storage model parameters. Close correspondence between values from the 53 parameter sets and the 1:1 line support the approximate relationship given by equation 14.

(F_{med}^{200}) is illustrated in Fig. 4, where values from the 53 parameter sets are shown to be in close agreement with equation 14 (Fig. 4, 1:1 line).

Discussion

F_{med}, *a new metric for transient storage*

Although equation 14 has not been derived directly from the transport equations, it is clear from Figs. 1, 3, and 4 and the associated discussion that F_{med} is some function of u, A_S, and α. This conclusion is further supported by the observation that values of F_{med} determined from numerical simulations change in response to changes in the model parameters, Q/A (u), A_S, and α. The goal of developing a transient storage metric that considers the interaction between these parameters has therefore been achieved and F_{med} is proposed as a new metric. Use of median travel time to define this new metric provides a clear physical interpretation: F_{med} is the fraction of median reach travel time due to storage. Stream reaches in which transient storage substantially affects the downstream transport of solute mass will have high values of F_{med}, whereas stream reaches in which storage has little effect on downstream transport will have low values of F_{med}.

For a given study reach, median travel time may be used to determine time in the main channel (t_{med}^{m}), time in the storage zone (t_{med}^{s}), and the fraction of travel time due to storage (F_{med}), where these quantities are evaluated using the actual length of the study reach (Table 4). Multiple experiments on the same study reach (e.g., the 20 WFWB parameter sets; see Appendix), or different study reaches with identical lengths may also be compared using the actual reach length.

For general comparisons of studies conducted at different scales, a standard distance of 200 m is proposed (F_{med}^{200}). Although this standard is subject to debate, it is based on the analysis presented in Table 2, and is in general agreement with the average reach length of the 53 parameter sets considered here ($\bar{L}$ = 180.6 m). Values of F_{med}^{200} are compared to several other metrics in Table 3. For each metric, a rank is also presented that corresponds to the rating a given parameter set would be given if the metric was

TABLE 4. Comparison of time in main channel and time in storage zone based on mean travel time, median travel time, and residence time metrics, where mean travel time is represented by volumetric residence time. T_{sto} = storage zone residence time, T_{str} = main channel residence time, mp = % of mass passing observation point prior to t_{vol}, t_{vol} = volumetric residence time, t_{vol}^m = volumetric residence time due to main channel, t_{vol}^s = volumetric residence time due to storage, zone, t_{med}^m = median travel time due to main channel, t_{med}^s = median travel time due to storage zone. All quantities evaluated at the published reach length. Abbreviations of reaches as in Table 1.

	Mean time in reach		Time in main channel (min)			Time in storage zone (min)		
Reach	t_{vol} (min)	mp (%)	t_{med}^m	t_{vol}^m	T_{str}	t_{med}^s	t_{vol}^s	T_{sto}
Cow-2	14.67	67.1	11.0	11.2	18.5	1.43	3.50	5.80
Cow-2b	17.28	65.2	12.1	12.2	13.9	2.39	5.09	5.80
Cow-3	6.97	68.4	6.2	6.1	8.3	0.34	0.85	1.17
Cow-4	10.85	64.1	8.7	8.7	11.1	1.09	2.11	2.69
Gall-Aut	248.89	87.1	41.1	44.5	273.2	2.62	204.44	1256.83
Gall-Spr	3.56	67.3	3.1	3.2	83.3	0.03	0.36	9.26
Gall-Sum	70.00	85.1	24.8	26.7	138.9	1.85	43.33	225.69
Gall-Win	15.20	81.4	8.9	8.9	33.3	0.39	6.33	23.81
HWC	50.71	69.8	19.2	20.4	28.4	6.57	30.29	42.12
LLM-1	62.17	75.8	42.8	46.6	416.7	1.68	15.54	138.89
LLM-2	349.07	78.9	62.8	69.8	208.3	10.16	279.25	833.33
LLM-3	258.92	67.3	64.8	74.0	83.3	53.91	184.95	208.33
LLM-4	364.57	83.1	173.0	182.3	833.3	13.05	182.28	833.33
SK86-1	11.90	97.3	8.4	8.4	555.6	0.02	3.50	231.48
SK86-2	159.04	89.4	105.5	105.0	833.3	0.61	54.09	429.55
SK86-3	23.59	99.5	8.6	8.4	833.3	0.01	15.20	1509.66
SK86-4	131.84	92.9	87.9	87.7	1111.1	0.30	44.10	558.35
SK88-1	10.89	96.1	5.9	6.0	166.7	0.05	4.88	135.14
SK88-2	24.26	91.6	14.7	15.0	166.7	0.30	9.26	102.88
SK88-3	117.15	84.3	34.9	35.0	166.7	0.78	82.17	391.50
SK88-4	56.26	93.4	40.6	40.6	555.6	0.23	15.63	213.68
Snake-1	30.52	94.1	26.2	26.2	666.7	0.09	4.30	109.29
Snake-2	23.86	68.8	23.1	23.0	333.3	0.10	0.84	12.12
Snake-3	20.82	77.7	11.4	11.6	33.3	1.07	9.18	26.32
Snake-4	12.94	92.1	10.1	10.1	166.7	0.09	2.81	46.30
Snake-5	45.02	74.2	38.5	38.4	83.3	0.68	6.59	14.29
Snake-7	8.34	78.3	8.0	8.0	333.3	0.01	0.37	15.43
Snake-8	21.85	85.5	21.0	21.0	333.3	0.05	0.87	13.77
Snake-9	63.02	80.4	48.1	48.0	166.7	0.51	15.01	52.08
Uvas-3	163.72	87.9	78.4	81.9	555.6	3.34	81.86	555.56
Uvas-4	154.45	94.5	72.5	77.2	1666.7	1.09	77.22	1666.67
Uvas-5	467.24	79.6	107.5	116.8	370.4	14.83	350.43	1111.11
WFWB	25.20	66.9	22.6	23.0	238.1	0.41	2.16	22.32
WFWB-1	23.51	61.6	21.1	21.5	111.1	0.76	1.97	10.19
WFWB-2	12.92	62.9	11.5	11.7	50.0	0.49	1.26	5.39
WFWB-3	5.95	56.6	5.5	5.5	10.5	0.31	0.41	0.78
WFWB-4	6.24	62.0	5.5	5.5	12.8	0.36	0.70	1.62
WFWB-5	13.13	67.4	11.5	11.7	66.7	0.37	1.46	8.33
WFWB-6	15.91	63.2	13.6	13.8	28.6	1.04	2.12	4.40
WFWB-7	13.27	66.4	11.4	11.7	58.8	0.47	1.60	8.09
WFWB-8	45.57	67.7	39.8	40.1	125.0	1.84	5.47	17.05
WFWB-9	35.00	67.7	31.0	31.4	200.0	0.85	3.59	22.86
WFWB-10	27.22	65.2	24.7	25.0	200.0	0.53	2.22	17.78
WFWB-11	22.87	65.1	20.6	21.0	250.0	0.36	1.87	22.22
WFWB-12	15.75	62.7	14.4	14.6	111.1	0.34	1.17	8.89
WFWB-13	46.67	61.2	41.1	41.4	71.4	2.95	5.27	9.09

TABLE 4. Continued.

Reach	Mean time in reach		Time in main channel (min)			Time in storage zone (min)		
	t_{vol} (min)	mp (%)	t_{med}^m	t_{vol}^m	T_{str}	t_{med}^s	t_{vol}^s	T_{sto}
WFWB-14	47.03	68.6	39.8	40.1	100.0	2.40	6.93	17.27
WFWB-15	44.84	65.5	39.6	40.1	166.7	1.57	4.74	19.70
WFWB-16	29.51	62.0	27.3	27.5	125.0	0.78	2.06	9.38
WFWB-17	29.85	66.3	27.1	27.5	333.3	0.41	2.40	29.17
WFWB-18	18.77	63.2	16.8	16.9	52.6	0.78	1.86	5.79
WFWB-19	21.17	62.4	19.2	19.4	100.0	0.62	1.73	8.89
WFWB-20	43.25	60.6	39.3	39.8	142.8	1.58	3.41	12.24

used to quantify storage zone importance. The highest-ranking parameter set based on F_{med}^{200} (reach 3 of Little Lost Man, LLM-3), for example, would be rated 1st, 3rd, 4th, 11th, or 14th, if storage zone importance were based on L_s, R_h, F_{mean}, q_s, or T_{sto}, respectively. Because F_{med} is the only metric that includes the interaction between u, α, and A_S, rankings for F_{med}^{200} differ from rankings based on other measures. Of the metrics considered, L_s, F_{mean}, and R_h are the most similar to F_{med}^{200} (i.e., 6 of the top 10 parameter sets based on F_{mean} also appear in the top 10 parameter sets based on F_{med}^{200}). For q_s and T_{sto}, the correspondence between the rankings is much more disparate (i.e., only 3 of the top 10 appear in the top 10 parameter sets based on F_{med}^{200}).

Although the standard distance of 200 m is proposed for analysis of the small streams presented here, this distance may not be appropriate for larger systems in which L = 200 m constitutes the near field and 1-dimensional analysis does not apply (Rutherford 1994). Additional analysis of F_{med} may be warranted when more studies of larger systems (e.g., Laenen and Bencala 2001) become available.

Mean, median, mass, and metrics

The advantages of using the new metric become evident when one considers the relationships between the mean, the median, and mass transport. As illustrated in Fig. 1, median travel time reflects the center of mass and is affected by α; mean travel time, in contrast, is unaffected by α and always occurs at a time when >50% of the mass has passed the observation point. The amount of mass passing the observation point prior to mean travel time (as represented t_{vol}) ranges from 56.6% (WFWB-3) to 99.5% (SK86-3) for the 53 parameter sets (Table 4).

The difference in mass transport for mean (>50% of mass) and median (= 50% of mass) travel time may be attributed to the storage zone by noting that the mean and median time in the main channel (t_{vol}^m, t_{med}^m) are comparable, whereas mean time in the storage zone (t_{vol}^s) exceeds median time in the storage zone (t_{med}^s) (Table 4). The longer storage zone times given by t_{vol}^s reflect the fact that mean travel time is independent of α, such that the entire volume of the storage zone contributes to travel time (2nd term on right-hand side of equation 9). In contrast, the median time in storage is limited by α, such that only a portion of the storage zone volume contributes to travel time. This limitation is evident from additional simulations that show t_{med}^s approaching t_{vol}^s as α approaches ∞ (for $uL/D > 50$ and $\alpha > 0.5$, $t_{med}^s \cong t_{vol}^s$) . From this analysis it is clear that mean travel time does not reflect the limiting effects of high velocity or low exchange coefficients (long L_s) on mass transport into the storage zone. F_{mean} (or A_S/A) therefore reflects the potential of storage to influence mass transport, whereas F_{med} reflects the degree to which the potential is realized. This distinction is especially important for reaches such as SK86-3, where >99% of the tracer mass passes through the study reach prior to mean travel time (Table 4). The relatively large A_S associated with this reach results in a large value of F_{mean} (64.4%, ranked 6 of 53; Table 3), but only a fraction of this potential is realized because of a long L_s (F_{med}^{200} = 0.17%, ranked 51 of 53; Table 3). Because of the large % of mass passing through the study reach prior to t_{vol} for most studies (Table 4), F_{mean} and A_S/A should be used with caution when quantifying mass-de-

pendent processes such as nutrient retention. F_{med} may be a more appropriate metric for this purpose because of the link between median travel time and the center of mass.

As with F_{mean}, values of T_{sto} may be misleading for parameter sets such as SK86-3. The general interpretation of this metric is that T_{sto} is equal to the time an average molecule spends in storage after it enters the storage zone. For the specific case of $L = L_s$, T_{sto} equals t_{vol}^{s} (and T_{str} equals t_{vol}^{m}); for $L_s > L$ the average molecule does not make its way into the storage zone and T_{sto} exceeds t_{vol}^{s} (and T_{str} exceeds t_{vol}^{m}) (Table 4). Characterizing transient storage based on the time the average molecule spends in the storage zone (T_{sto}) may therefore overestimate the importance of transient storage, as the average molecule never reaches the storage zone over the spatial scale studied. Parameter set SK86-3, for example, has a large value of T_{sto} (ranked 2 of 53; Table 3) that is of little consequence for mass transport (F_{med}^{200} = 0.17%, ranked 51 of 53; Table 3) because of the long distance travelled (L_s = 4170.7 m). Further, 2 parameters sets (e.g., LLM-2 and LLM- 4) may have identical values of T_{sto}, yet have vastly different amounts of mass entering the storage zone (F_{med}^{200} = 62.4% and 9.1% for LLM-2 and LLM-4, respectively; Table 3). This disconnect between mass transport and T_{sto} is a result of the contrasting effects of α on travel time and storage zone residence time. As discussed above, the median time in storage increases with α (t_{med}^{s} approaches t_{vol}^{s} as α approaches ∞), indicating an increase in mass exchange. Values of T_{sto}, in contrast, decrease as α increases.

In conclusion, F_{med} is proposed as an effective way to quantify the effects of transient storage in the context of whole-stream mass transport. Although metrics such as T_{sto} and A_S/A quantify the storage process, they are theoretically related to mean travel time, a quantity that is markedly influenced by extreme values of the travel time distribution. The net effect of this relationship is that T_{sto} and A_S/A do not quantify storage relative to the total amount of mass that is transported downstream. Long residence times within the storage zone as measured by T_{sto} (and large storage zone volumes as measured by A_S/A), for example, may be relatively unimportant if only a small fraction of the total solute mass makes its way into the storage zone. This distinction is especially important for point sources of solutes such as experimental nutrient additions (e.g., Mulholland et al. 1997), accidental spills, and wastewater treatment plant effluent. In these cases, solute mass directly enters the main channel in a manner that is analogous to a tracer injection. Measures such as F_{med} that quantify the movement of tracer into the storage zone, relative to mass transport, are therefore of paramount importance when comparing the storage characteristics of different streams and rivers.

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Appendix

Parameter sets used to evaluate median travel time due to storage. A = main channel cross-sectional area, A_S = storage zone cross-sectional area, D = dispersion coefficient, L = reach length, Pe = Peclet number ($= uL/D$), Q_0 = flow rate at top of reach, q_L = lateral inflow rate, u = velocity, and α = storage zone exchange coefficient. Abbreviations of reaches as in Table 1.

APPENDIX. Parameter sets used to evaluate median travel time due to storage. A = main channel cross-sectional area, A_S = storage zone cross-sectional area, D = dispersion coefficient, L = reach length, Pe = Peclet number (= uL/D), Q_O = flow rate at top of each, q_L = lateral inflow rate, u = velocity, and α = storage zone exchange coefficient. Abbreviations of reaches as in Table 1.

	OTIS input values								
Reach	L (m)	D (m^2/s)	α (/s)	A_S (m^2)	A (m^2)	Q_O (m^3/s)	q_L (m^3/s-m)	u (m/s)	Pe
Cow-2	105	0.30	9.00×10^{-4}	0.120	0.383	6.00×10^{-2}	0.0	0.157	54.8
Cow-2b	105	0.20	1.20×10^{-3}	0.160	0.383	5.50×10^{-2}	0.0	0.144	75.4
Cow-3	100	0.05	2.00×10^{-3}	0.080	0.572	1.56×10^{-1}	0.0	0.273	545.5
Cow-4	100	0.20	1.50×10^{-3}	0.350	1.446	2.76×10^{-1}	0.0	0.191	95.4
Gall-Aut	40	0.05	6.10×10^{-5}	0.230	0.050	7.50×10^{-4}	0.0	0.015	12.0
Gall-Spr	40	0.40	2.00×10^{-4}	0.040	0.360	7.50×10^{-2}	0.0	0.208	20.8
Gall-Sum	40	0.08	1.20×10^{-4}	0.130	0.080	2.00×10^{-3}	0.0	0.025	12.5
Gall-Win	57	0.05	5.00×10^{-4}	0.100	0.140	1.50×10^{-2}	0.0	0.107	122.1
HWC	78	0.20	5.87×10^{-4}	0.129	0.087	4.90×10^{-3}	1.63×10^{-5}	0.064	24.8
LLM-1	57	0.10	4.00×10^{-5}	0.120	0.360	7.25×10^{-3}	2.98×10^{-6}	0.020	11.6
LLM-2	63	0.10	8.00×10^{-5}	2.000	0.500	7.42×10^{-3}	3.18×10^{-6}	0.015	9.5
LLM-3	57	0.10	2.00×10^{-4}	1.500	0.600	7.62×10^{-3}	2.98×10^{-6}	0.013	7.3
LLM-4	142	0.10	2.00×10^{-5}	0.600	0.600	7.79×10^{-3}	0.0	0.013	18.4
SK86-1	26	0.02	3.00×10^{-5}	0.050	0.120	6.19×10^{-3}	0.0	0.052	67.1
SK86-2	458	0.02	2.00×10^{-5}	0.050	0.097	6.19×10^{-3}	3.78×10^{-6}	0.073	1665.7
SK86-3	42	0.02	2.00×10^{-5}	0.250	0.138	7.92×10^{-3}	1.71×10^{-4}	0.083	175.2
SK86-4	422	0.02	1.50×10^{-5}	0.100	0.199	1.51×10^{-2}	4.03×10^{-6}	0.080	1691.2
SK88-1	24	0.05	1.00×10^{-4}	0.150	0.185	1.23×10^{-2}	0.0	0.066	31.9
SK88-2	46	0.05	1.00×10^{-4}	0.150	0.243	1.23×10^{-2}	5.10×10^{-6}	0.051	47.0
SK88-3	181	0.05	1.00×10^{-4}	0.350	0.149	1.25×10^{-2}	3.53×10^{-6}	0.086	312.2
SK88-4	247	0.05	3.00×10^{-5}	0.050	0.130	1.32×10^{-2}	0.0	0.101	500.5
Snake-1	628	0.75	2.50×10^{-5}	0.100	0.610	2.24×10^{-1}	6.20×10^{-5}	0.399	334.2
Snake-2	737	0.60	5.00×10^{-5}	0.020	0.550	2.63×10^{-1}	8.30×10^{-5}	0.534	655.5
Snake-3	122	0.50	5.00×10^{-4}	1.500	1.900	3.24×10^{-1}	1.31×10^{-4}	0.175	42.6
Snake-4	118	0.20	1.00×10^{-4}	0.500	1.800	3.40×10^{-1}	1.61×10^{-4}	0.194	114.6
Snake-5	1240	0.20	2.00×10^{-4}	0.120	0.700	3.59×10^{-1}	2.80×10^{-5}	0.538	3334.4
Snake-7	279	0.20	5.00×10^{-5}	0.050	1.080	6.06×10^{-1}	1.76×10^{-4}	0.584	814.2
Snake-8	697	0.20	5.00×10^{-5}	0.050	1.210	6.55×10^{-1}	4.30×10^{-5}	0.554	1929.4
Snake-9	1342	0.20	1.00×10^{-4}	0.500	1.600	6.85×10^{-1}	9.00×10^{-5}	0.466	3125.6
Uvas-3	176	0.24	3.00×10^{-5}	0.360	0.360	1.25×10^{-2}	4.55×10^{-6}	0.036	26.3
Uvas-4	152	0.31	1.00×10^{-5}	0.410	0.410	1.33×10^{-2}	1.97×10^{-6}	0.033	16.1
Uvas-5	186	0.40	4.50 ± 10^{-5}	1.560	0.520	1.36×10^{-2}	2.15×10^{-6}	0.027	12.3
WFWB	62	0.05	7.00×10^{-5}	0.009	0.096	4.20×10^{-3}	3.39×10^{-6}	0.045	55.6
WFWB-1	70	0.09	1.50×10^{-4}	0.011	0.120	6.50×10^{-3}	0.0	0.054	42.1
WFWB-2	70	0.15	3.33×10^{-4}	0.014	0.130	1.30×10^{-2}	0.0	0.100	46.7
WFWB-3	70	0.31	1.58×10^{-3}	0.014	0.190	4.00×10^{-2}	0.0	0.211	47.5
WFWB-4	70	0.25	1.30×10^{-3}	0.024	0.190	4.00×10^{-2}	0.0	0.211	58.9
WFWB-5	70	0.12	2.50×10^{-4}	0.015	0.120	1.20×10^{-2}	0.0	0.100	58.3
WFWB-6	70	0.09	5.83×10^{-4}	0.020	0.130	1.10×10^{-2}	0.0	0.085	65.8
WFWB-7	70	0.17	2.83×10^{-4}	0.022	0.160	1.60×10^{-2}	0.0	0.100	41.2
WFWB-8	70	0.02	1.33×10^{-4}	0.015	0.110	3.20×10^{-3}	0.0	0.029	101.8
WFWB-9	70	0.04	8.33×10^{-5}	0.008	0.070	2.60×10^{-3}	0.0	0.037	65.0
WFWB-10	70	0.05	8.33×10^{-5}	0.008	0.090	4.20×10^{-3}	0.0	0.047	65.3
WFWB-11	70	0.09	6.67×10^{-5}	0.008	0.090	5.00×10^{-3}	0.0	0.056	43.2
WFWB-12	70	0.11	1.50×10^{-4}	0.012	0.150	1.20×10^{-2}	0.0	0.080	50.9
WFWB-13	70	0.02	2.33×10^{-4}	0.014	0.110	3.10×10^{-3}	0.0	0.028	98.6
WFWB-14	70	0.02	1.67×10^{-4}	0.019	0.110	3.20×10^{-3}	0.0	0.029	101.8
WFWB-15	70	0.03	1.00×10^{-4}	0.013	0.110	3.20×10^{-3}	0.0	0.029	67.9
WFWB-16	70	0.03	1.33×10^{-4}	0.006	0.080	3.40×10^{-3}	0.0	0.042	99.2

APPENDIX. Continued.

	OTIS input values								
Reach	L (m)	D (m^2/s)	α (/s)	A_S (m^2)	A (m^2)	Q_O (m^3/s)	q_L (m^3/s-m)	u (m/s)	Pe
WFWB-17	70	0.05	5.00×10^{-5}	0.007	0.080	3.40×10^{-3}	0.0	0.042	59.5
WFWB-18	70	0.06	3.17×10^{-4}	0.011	0.100	6.90×10^{-3}	0.0	0.069	80.5
WFWB-19	70	0.07	1.67×10^{-4}	0.008	0.090	5.40×10^{-3}	0.0	0.060	60.0
WFWB-20	70	0.03	1.17×10^{-4}	0.012	0.140	4.10×10^{-3}	0.0	0.029	68.3